

Chemical Engineering Lab – I
Experiment 1

Flow Through Fluidized Bed and Packed Bed (Water)

Submitted by

Section: P-1 Group: 2

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Experiment 1(a): Flow Through Fluidized Bed (Water)

1. Aim

Study of hydrodynamic and bed characteristics for flow through fluidized beds.

2. Objective

- To determine the minimum fluidization velocity experimentally as well as theoretically.
- To study the bed expansion characteristics of the fluidized bed (plot $\log N_{Re}$ vs. porosity, and pressure drop vs. N_{Re}).

3. Formulae Used

Quantity	Formula
Superficial velocity	$U = \frac{Q}{A}$
Bed expansion (porosity relation)	$Z_2 = Z_1 \frac{(1 - \varepsilon_1)}{(1 - \varepsilon_2)}$
Modified Reynolds Number	$N_{Re} = \frac{\rho U d_p}{\mu(1 - \varepsilon_2)}$
Pressure drop (manometer)	$\Delta P = \Delta h g (\rho_{Hg} - \rho)$
Archimedes Number	$Ar = \frac{d_p^3 \rho (\rho_p - \rho) g}{\mu^2}$
Minimum fluidization Reynolds No. (Wen & Yu)	$Re_{mf} = \sqrt{1135.7 + 0.0408 Ar} - 33.7$
Minimum fluidization velocity	$U_{mf} = \frac{Re_{mf} \mu}{\rho d_p}$
Theoretical ΔP at incipient fluidization	$\Delta P = Z_1(1 - \varepsilon_1)(\rho_p - \rho)g$

4. Data / Specifications

Property	Value
Particle size, d_p	8 mm
Inside diameter of column, D	0.055 m
Cross-sectional area, A	0.002375 m ²
Initial (static) bed height, Z_1	0.097 m (9.7 cm)
Static bed porosity, ε_1	0.6
Density of water, ρ	1000 kg/m ³
Viscosity of water, μ	0.798×10^{-3} kg/m-s
Manometer fluid, ρ_{Hg}	13,534 kg/m ³
Particle (packing) density, ρ_p ¹	2500 kg/m ³

5. Observations

S.No.	Flow rate, Q (LPM)	Flow rate, Q (m ³ /s)	Bed height, Z_2 (cm)
1	4	6.6×10^{-5}	10.5
2	6	1.0×10^{-4}	12.5
3	8	1.33×10^{-4}	15.0
4	10	1.66×10^{-4}	18.8
5	12	2.0×10^{-4}	22.4

S.No.	Flow rate, Q (LPM)	Manometer reading, Δh (mm)
1	4	20
2	6	31
3	8	45
4	10	63
5	12	78

6. Sample Calculation (S.No. 1, $Q = 4$ LPM)

Superficial velocity:

$$U = \frac{Q}{A} = \frac{6.6 \times 10^{-5}}{0.002375} = 0.02779 \text{ m/s}$$

Pressure drop:

$$\Delta P = \Delta h g (\rho_{Hg} - \rho) = 0.02 \times 9.81 \times (13534 - 1000) = 2459.17 \text{ Pa}$$

¹Particle density not listed in the current manual; sourced from prior group reports (Groups 15 and 5, Exp. 1) for the same glass-ring packing, where it appears consistently.

Fluidized-bed porosity:

$$\varepsilon_2 = 1 - \frac{Z_1(1 - \varepsilon_1)}{Z_2} = 1 - \frac{0.097 \times 0.4}{0.105} = 0.6305$$

Modified Reynolds Number:

$$N_{Re} = \frac{\rho U d_p}{\mu(1 - \varepsilon_2)} = \frac{1000 \times 0.02779 \times 0.008}{0.798 \times 10^{-3} \times 0.3695} = 753.92$$

Theoretical calculation (minimum fluidization):

$$Ar = \frac{d_p^3 \rho(\rho_p - \rho)g}{\mu^2} = 1.183 \times 10^7$$

$$Re_{mf} = \sqrt{1135.7 + 0.0408 Ar} - 33.7 = 661.9$$

$$U_{mf} = \frac{Re_{mf} \mu}{\rho d_p} = 0.0660 \text{ m/s}$$

$$\Delta P_{theoretical} = Z_1(1 - \varepsilon_1)(\rho_p - \rho)g = 570.9 \text{ Pa}$$

7. Calculation Table

S.No.	Q (m ³ /s)	U (m/s)	ΔP (Pa)	ε_2	N_{Re}
1	6.6×10^{-5}	0.0278	2459.17	0.6305	753.92
2	1.0×10^{-4}	0.0421	3811.71	0.6896	1359.88
3	1.33×10^{-4}	0.0560	5533.13	0.7413	2170.37
4	1.66×10^{-4}	0.0699	7746.39	0.7936	3395.14
5	2.0×10^{-4}	0.0842	9590.77	0.8268	4873.82

Experiment 1(b): Flow Through Packed Bed (Water)

1. Aim

- To plot Modified Reynolds No. ($N_{Re,p}$) vs. Modified friction factor (f_p) on a log-log graph.
- To verify Ergun's equation.

2. Formulae Used

Quantity	Formula
Particle diameter	$D_p = \frac{6V_p}{S_p}$
Superficial velocity	$V_0 = \frac{Q}{A}$
Pressure drop (manometer)	$\Delta P = \Delta h g (\rho_{Hg} - \rho)$
Modified friction factor	$f_p = \frac{\Delta P}{L} \frac{D_p}{\rho V_0^2} \left(\frac{\varepsilon^3}{1 - \varepsilon} \right)$
Modified Reynolds Number	$Re_p = \frac{\rho V_0 D_p}{\mu(1 - \varepsilon)}$
Ergun's equation (theoretical)	$\frac{\Delta P}{L} = \frac{150\mu V_0(1 - \varepsilon)^2}{D_p^2 \varepsilon^3} + \frac{1.75\rho V_0^2(1 - \varepsilon)}{D_p \varepsilon^3}$

3. Data / Specifications

Property	Value
Inside diameter of tube, D	0.050 m
Cross-sectional area, A	$1.9635 \times 10^{-3} \text{ m}^2$
Volume of each particle, V_p	$1.57 \times 10^{-6} \text{ m}^3$
Surface area of particle, S_p	$6.28 \times 10^{-4} \text{ m}^2$
Particle diameter, $D_p = 6V_p/S_p$	0.015 m
Porosity of bed, ε	0.3
Effective length of packing, L	0.75 m
Density of water, ρ	1000 kg/m ³
Viscosity of water, μ	$0.798 \times 10^{-3} \text{ kg/m-s}$
Manometer fluid, ρ_{Hg}	13,534 kg/m ³

4. Observations

S.No.	Flow rate, Q (LPM)	Flow rate, Q (m ³ /s)	Manometer reading, Δh (mm)
1	4	6.6×10^{-5}	10
2	6	1.0×10^{-4}	16
3	8	1.33×10^{-4}	27
4	10	1.66×10^{-4}	38
5	12	2.0×10^{-4}	52

5. Sample Calculation (S.No. 1, $Q = 4$ LPM)

Particle diameter:

$$D_p = \frac{6V_p}{S_p} = \frac{6 \times 1.57 \times 10^{-6}}{6.28 \times 10^{-4}} = 0.015 \text{ m}$$

Superficial velocity:

$$V_0 = \frac{Q}{A} = \frac{6.6 \times 10^{-5}}{1.9635 \times 10^{-3}} = 0.03361 \text{ m/s}$$

Pressure drop:

$$\Delta P = \Delta h g (\rho_{Hg} - \rho) = 0.01 \times 9.81 \times (13534 - 1000) = 1229.59 \text{ Pa}, \quad \frac{\Delta P}{L} = 1639.45 \text{ Pa/m}$$

Modified friction factor:

$$f_p = \frac{\Delta P}{L} \cdot \frac{D_p}{\rho V_0^2} \cdot \frac{\varepsilon^3}{1 - \varepsilon} = 1639.45 \times \frac{0.015}{1000 \times (0.03361)^2} \times \frac{0.027}{0.7} = 0.8395$$

Modified Reynolds Number:

$$Re_p = \frac{\rho V_0 D_p}{\mu(1 - \varepsilon)} = \frac{1000 \times 0.03361 \times 0.015}{0.798 \times 10^{-3} \times 0.7} = 902.62$$

Theoretical (Ergun) pressure gradient:

$$\left(\frac{\Delta P}{L} \right)_{theo} = \frac{150 \mu V_0 (1 - \varepsilon)^2}{D_p^2 \varepsilon^3} + \frac{1.75 \rho V_0^2 (1 - \varepsilon)}{D_p \varepsilon^3} = 3742.04 \text{ Pa/m}$$

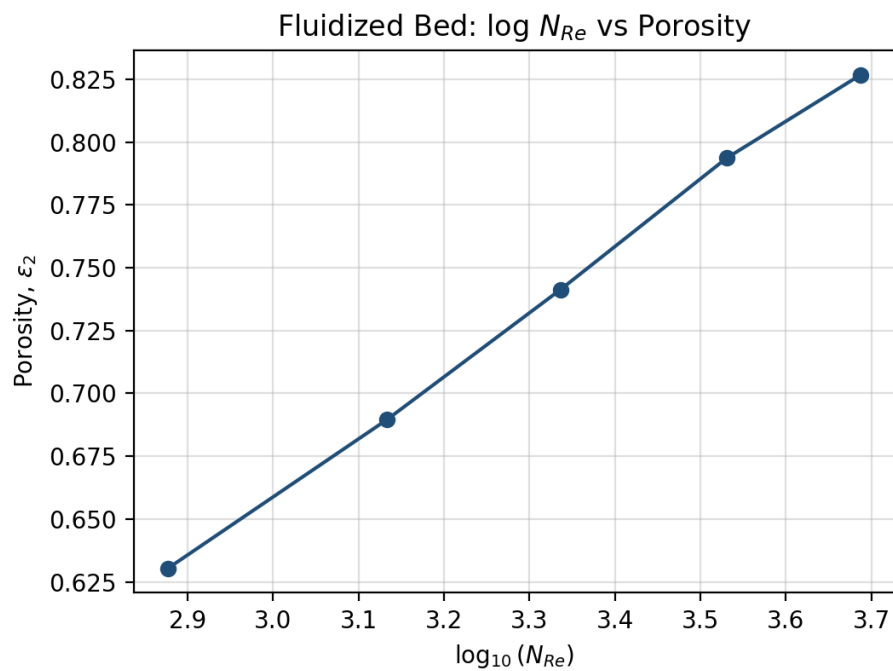
6. Calculation Table

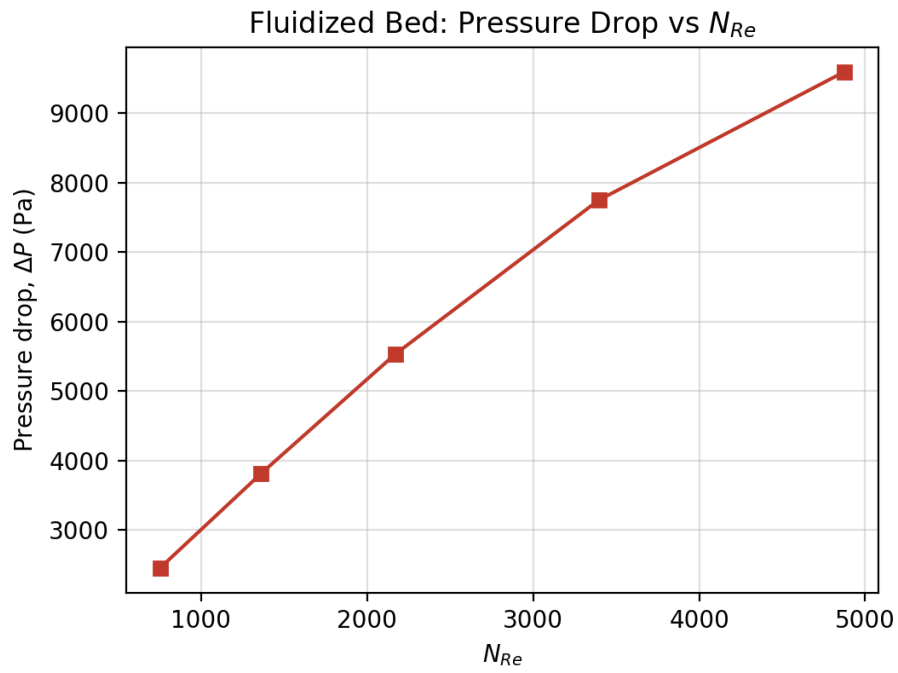
S.No.	V_0 (m/s)	$\Delta P/L$ (Pa/m)	Re_p	$(\Delta P/L)_{theo}$ (Pa/m)	f_p
1	0.0336	1639.45	902.62	3742.04	0.8395
2	0.0509	2623.12	1367.60	8337.23	0.5851
3	0.0677	4426.51	1818.91	14531.91	0.5582
4	0.0845	6229.90	2270.22	22435.34	0.5043
5	0.1019	8525.13	2735.21	32365.48	0.4754

7. Results

7.1 Fluidized Bed

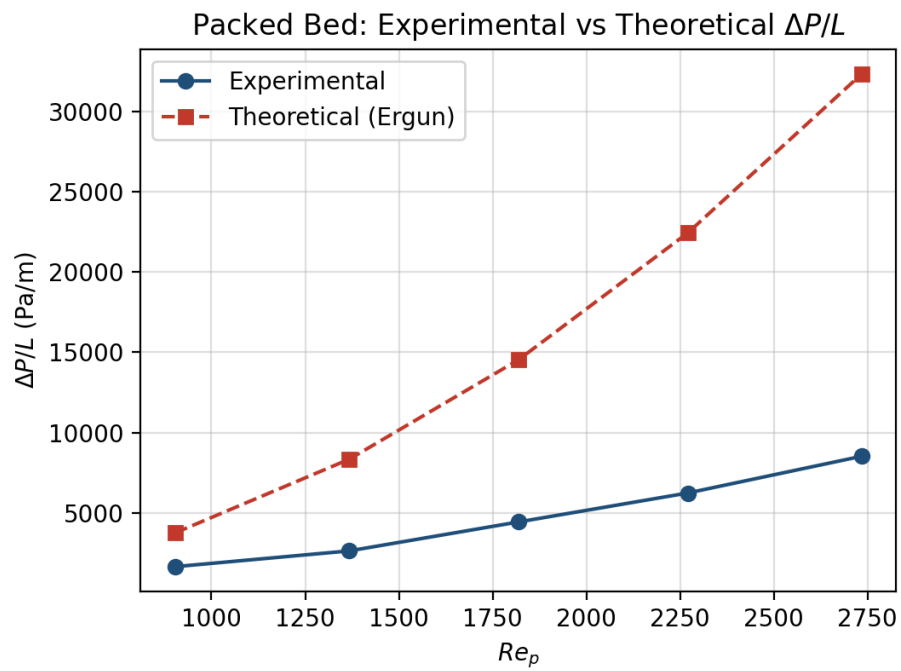
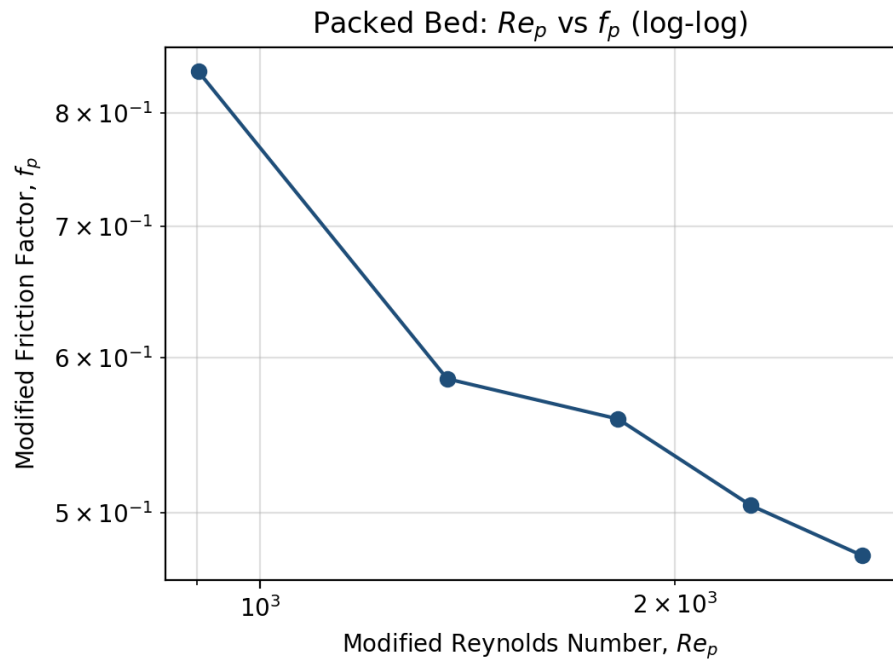
Quantity	Value
Minimum fluidization velocity, U_{mf} (theoretical)	0.0660 m/s
Theoretical ΔP at incipient fluidization	570.9 Pa
Range of experimental N_{Re}	753.9 – 4873.8
Range of experimental porosity, ε_2	0.630 – 0.827
Range of experimental ΔP	2459.2 – 9590.8 Pa





7.2 Packed Bed

Quantity	Value
Range of experimental Re_p	902.6 – 2735.2
Range of experimental f_p	0.475 – 0.840
Range of $\Delta P/L$ (experimental)	1639.4 – 8525.1 Pa/m
Range of $\Delta P/L$ (theoretical, Ergun)	3742.0 – 32365.5 Pa/m



10. Precautions

1. Steady-state flow was ensured at each setting before noting readings.
2. Manometer readings were taken only after the mercury column stabilized, avoiding parallax error.
3. The bed height was measured at the same reference point on the column for every observation.
4. Flow rate was increased gradually to avoid disturbing the packing or causing sudden surges.
5. Air bubbles were avoided in the manometer lines to prevent erroneous Δh readings.

References

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2. Geankoplis, Christie J., *Transport Processes and Unit Operations*, 4th ed., New Jersey: Prentice Hall, 2003.
3. Ergun, Sabri, “Fluid Flow through Packed Columns,” *Chemical Engineering Progress*, Vol. 48, No. 2, 89–94, 1952.
4. Warren, L. McCabe, Smith, J. C., and Harriott, P., *Unit Operations of Chemical Engineering*, 6th ed., McGraw Hill, New Delhi, India, 2000.
5. Kunii, D., Levenspiel, O., *Fluidization Engineering*, Robert E. Krieger Publishing Company, Huntington, New York, 1977.